Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Blevins, Linda G. Ph.D.		
Project Title:	Carbon Monoxide and Soot Formation in Inverse Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lgblevi@sandia.gov	**Fax:**	FY 925-294-2276
PI Organization Type:	GOVERNMENT	**Phone:**	925-294-4811
Organization Name:	Sandia National Laboratories		
PI Address 1:	Combustion Research Facility		
PI Address 2:	P.O. Box 969 MS 9052		
PI Web Page:	http://www.science.doe.gov/		
City:	Livermore	**State:**	CA
Zip Code:	94551-0969	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	02/01/1999	**End Date:**	12/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Feikema, Douglas A	**Contact Phone:**	216-433-5707
Contact Email:	Douglas.A.Feikema@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulholland, George W (National Institute of Standards and Technology) Davis, Ronald W (National Institute of Standards and Technology) Fernandez-Pello, Carlos (University of California Berkeley)		
Grant/Contract No.:	C-32100N		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	Experimental work is combined with numerical modeling. The project tasks are: 1. to stabilize methane and ethene inverse flames in the 2.2-second drop tower, 2. to collect and measure the amounts of carbone monoxide (CO) and soot produced by these flames, 3. to analyze the collected soot for size, shape, hydrogen/carbon content, and organic/graphitic makeup, 4. to develop a fiber-optic-coupled CO sensor for monitoring flames during drops, and 5. to use numerical modeling to predict the pathways followed by CO and soot in these flames. Measured CO and soot information will be combined with local computed flow pathways to illuminate the ways that CO and soot are formed during underventilated combustion. The objective of this project is to study CO and soot formation during the dangerous underventilated burning period characteristic of certain fires. Under-ventilated burning is obtained in the present study using an inverse diffusion flame established at the interface of a round air jet and an annular fuel stream. Microgravity is used to ensure that a wide range of stable, laminar, inverse-flame operating conditions can be achieved. The project will provide fundamental knowledge aimed to improve the CO and soot predictive capabilities of existing fire-behavior computer models. In addition, an understanding of noxious gas formation and flame soot signatures during underventilated fires in spacecraft will be obtained. Finally, the existing knowledge of pollutant formation in practical, staged combustors will be improved.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)